PDE-RSI benchmark problems

Generated by `scripts/write_problem_latex.py` from `src/pde_rsi/tasks.py`. Every problem has a manufactured exact solution u; the forcing f is obtained by substituting u into the equation. Agents never see u, the profile parameters (A, β or p, the spatial profile), or the closed form of f; solver programs receive f and the initial data only as black-box functions.

Caputo derivatives. For $0 < \alpha < 1$, with history from $t = 0$,

$$ {}_0^C D_t^\alpha u(x,t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t (t-s)^{-\alpha}\, \partial_s u(x,s)\, ds, \qquad {}_0^C D_t^\alpha t^p = \frac{\Gamma(p+1)}{\Gamma(p+1-\alpha)}\, t^{p-\alpha}. $$

A variable order $\alpha(t)$ or $\alpha(x,t)$ is evaluated at the current point: ${}_0^C D_t^{\alpha(x,t)} u = \frac{1}{\Gamma(1-\alpha(x,t))} \int_0^t (t-s)^{-\alpha(x,t)} \partial_s u(x,s)\, ds$.

Evaluation. A case passes when the space–time and final-time relative L^2 errors are both at most 2×10^{-3}, the initial and boundary errors are at most 10^{-10}, and the solve fits in 15 s and 1 GB. Errors use linear interpolation onto uniform nodes and the trapezoidal rule: 512×500 for 1D families (Poisson: 512 points; FODE: 500 times) and $128 \times 128 \times 100$ for the 2D wave equation.

1 One-dimensional benchmark (48 public, 24 reserved cases)

All cases use $x \in [0, L]$, $u(0,t) = u(L,t) = 0$ and the exact solution

$$ u(x,t) = A\, s(x)\, \big(1 + t^\beta\big) \quad \text{(Poisson: } u(x) = A\, s(x)\text{)}, \qquad u(x,0) = A\, s(x), $$

with the spatial profile fixed by the split:

$$ s(x) = \begin{cases} \sin(\pi x/L) & \text{train,} \\ 4\,\dfrac{x}{L}\Big(1 - \dfrac{x}{L}\Big) & \text{validation,} \\ \sin(\pi x/L) + \frac{1}{4}\sin(3\pi x/L) & \text{reserved.} \end{cases} $$

Write $g(t) = 1 + t^\beta$.

1.1 Poisson

$$ -D\, u_{xx} = f(x), \quad x \in (0, L) $$
$$ f = -D\, A\, s''(x) $$

Case	Split	L	D	A	Profile
`poisson-g0-v0`	train	1	0.05	0.3	sine
`poisson-g0-v1`	train	1.2	0.1	0.4	sine
`poisson-g0-v2`	train	1	0.15	0.5	sine
`poisson-g0-v3`	train	1.2	0.2	0.6	sine
`poisson-g1-v0`	validation	1	0.05	0.3	quadratic
`poisson-g1-v1`	validation	1.2	0.1	0.4	quadratic
`poisson-g1-v2`	validation	1	0.15	0.5	quadratic
`poisson-g1-v3`	validation	1.2	0.2	0.6	quadratic

Reserved cases: 4 (parameters kept evaluator-only).

1.2 Heat

$$u_t - D\,u_{xx} = f(x,t), \quad x \in (0,L),\ t \in (0,T]$$
$$f = A\big[\beta t^{\beta-1} s(x) - D\,g(t)\,s''(x)\big]$$

Case	Split	L	T	D	A	β	Profile
`heat-g0-v0`	train	1	0.25	0.05	0.3	1	sine
`heat-g0-v1`	train	1.2	0.25	0.1	0.4	1	sine
`heat-g0-v2`	train	1	0.35	0.15	0.5	1	sine
`heat-g0-v3`	train	1.2	0.35	0.2	0.6	1	sine
`heat-g1-v0`	validation	1	0.25	0.05	0.3	2	quadratic
`heat-g1-v1`	validation	1.2	0.25	0.1	0.4	2	quadratic
`heat-g1-v2`	validation	1	0.35	0.15	0.5	2	quadratic
`heat-g1-v3`	validation	1.2	0.35	0.2	0.6	2	quadratic

Reserved cases: 4 (parameters kept evaluator-only).

1.3 Convection–diffusion

$$u_t + v\,u_x - D\,u_{xx} = f(x,t)$$
$$f = A\big[\beta t^{\beta-1} s - D\,g\,s'' + v\,g\,s'\big]$$

Case	Split	L	T	D	v	A	β	Profile
`convection_diffusion-g0-v0`	train	1	0.25	0.05	0.25	0.3	1	sine
`convection_diffusion-g0-v1`	train	1.2	0.25	0.1	0.35	0.4	1	sine
`convection_diffusion-g0-v2`	train	1	0.35	0.15	0.45	0.5	1	sine
`convection_diffusion-g0-v3`	train	1.2	0.35	0.2	0.55	0.6	1	sine
`convection_diffusion-g1-v0`	validation	1	0.25	0.05	0.25	0.3	2	quadratic
`convection_diffusion-g1-v1`	validation	1.2	0.25	0.1	0.35	0.4	2	quadratic
`convection_diffusion-g1-v2`	validation	1	0.35	0.15	0.45	0.5	2	quadratic
`convection_diffusion-g1-v3`	validation	1.2	0.35	0.2	0.55	0.6	2	quadratic

Reserved cases: 4 (parameters kept evaluator-only).

1.4 Viscous Burgers

$$u_t + u\,u_x - D\,u_{xx} = f(x,t)$$
$$f = A\big[\beta t^{\beta-1} s - D\,g\,s''\big] + A^2 g^2\,s\,s'$$

Case	Split	L	T	D	A	β	Profile
`burgers-g0-v0`	train	1	0.25	0.05	0.3	1	sine
`burgers-g0-v1`	train	1.2	0.25	0.1	0.4	1	sine
`burgers-g0-v2`	train	1	0.35	0.15	0.5	1	sine
`burgers-g0-v3`	train	1.2	0.35	0.2	0.6	1	sine
`burgers-g1-v0`	validation	1	0.25	0.05	0.3	2	quadratic
`burgers-g1-v1`	validation	1.2	0.25	0.1	0.4	2	quadratic
`burgers-g1-v2`	validation	1	0.35	0.15	0.5	2	quadratic
`burgers-g1-v3`	validation	1.2	0.35	0.2	0.6	2	quadratic

Reserved cases: 4 (parameters kept evaluator-only).

1.5 Caputo subdiffusion

$$
{}_0^C D_t^{\alpha} u - D\, u_{xx} = f(x,t), \quad 0 < \alpha < 1
$$

$$
f = A\Big[\frac{\Gamma(\beta+1)}{\Gamma(\beta+1-\alpha)} t^{\beta-\alpha} s - D\, g\, s''\Big]
$$

Case	Split	L	T	D	A	β	α	Profile
`caputo-g0-v0`	train	1	0.25	0.05	0.3	2	0.4	sine
`caputo-g0-v1`	train	1.2	0.25	0.1	0.4	2	0.55	sine
`caputo-g0-v2`	train	1	0.35	0.15	0.5	2	0.7	sine
`caputo-g0-v3`	train	1.2	0.35	0.2	0.6	2	0.85	sine
`caputo-g1-v0`	validation	1	0.25	0.05	0.3	0.8	0.4	quadratic
`caputo-g1-v1`	validation	1.2	0.25	0.1	0.4	0.8	0.55	quadratic
`caputo-g1-v2`	validation	1	0.35	0.15	0.5	0.8	0.7	quadratic
`caputo-g1-v3`	validation	1.2	0.35	0.2	0.6	0.8	0.85	quadratic

Reserved cases: 4 (parameters kept evaluator-only).

1.6 Multi-term Caputo diffusion

$$
\sum_{j=1}^{2} w_j\, {}_0^C D_t^{\alpha_j} u - D\, u_{xx} = f(x,t)
$$

$$
f = A\Big[\sum_j w_j \frac{\Gamma(\beta+1)}{\Gamma(\beta+1-\alpha_j)} t^{\beta-\alpha_j} s - D\, g\, s''\Big]
$$

Case	Split	L	T	D	A	β	(α_1, α_2)	(w_1, w_2)	Profile
`multi_term-g0-v0`	train	1	0.25	0.05	0.3	2	(0.3, 0.7)	(0.4, 1)	sine
`multi_term-g0-v1`	train	1.2	0.25	0.1	0.4	2	(0.35, 0.75)	(0.4, 1)	sine
`multi_term-g0-v2`	train	1	0.35	0.15	0.5	2	(0.4, 0.8)	(0.4, 1)	sine
`multi_term-g0-v3`	train	1.2	0.35	0.2	0.6	2	(0.45, 0.85)	(0.4, 1)	sine
`multi_term-g1-v0`	validation	1	0.25	0.05	0.3	0.8	(0.3, 0.7)	(0.4, 1)	quadratic
`multi_term-g1-v1`	validation	1.2	0.25	0.1	0.4	0.8	(0.35, 0.75)	(0.4, 1)	quadratic
`multi_term-g1-v2`	validation	1	0.35	0.15	0.5	0.8	(0.4, 0.8)	(0.4, 1)	quadratic
`multi_term-g1-v3`	validation	1.2	0.35	0.2	0.6	0.8	(0.45, 0.85)	(0.4, 1)	quadratic

Reserved cases: 4 (parameters kept evaluator-only).

2 Two-dimensional wave equation (8 public, 4 reserved cases)

$$
\begin{aligned}
&u_{tt} - D\,(u_{xx} + u_{yy}) = f(x,y,t), \qquad (x,y) \in (0,L)^2,\ t \in (0,T],\\
&u = 0 \text{ on } \partial(0,L)^2, \qquad u(x,y,0) = A\, s(x) s(y), \qquad u_t(x,y,0) = 0,\\
&u(x,y,t) = A\, s(x)\, s(y)\, \big(\cos \omega t + t^{\beta}\big), \qquad \omega = \sqrt{D}\, \pi \sqrt{2}/L,\\
&f = A\Big[\big(-\omega^2 \cos \omega t + \beta(\beta-1) t^{\beta-2}\big) S - D\big(\cos \omega t + t^{\beta}\big) \Delta S\Big],\\
&S(x,y) = s(x)\, s(y),
\end{aligned}
$$

with s as in the 1D benchmark (train: sine, $\beta = 2.5$; validation: quadratic, $\beta = 1.5$, so $u_{tt} \sim t^{-1/2}$ is singular at $t = 0$; reserved: mixed, $\beta = 1.25$). $D = c^2$ is the squared wave speed. Frequencies ω: `wave2d-g0-v0`: 2.2214, `wave2d-g0-v1`: 2.7768, `wave2d-g0-v2`: 4.4429, `wave2d-g0-v3`: 4.6280, `wave2d-g1-v0`: 2.2214, `wave2d-g1-v1`: 2.7768, `wave2d-g1-v2`: 4.4429, `wave2d-g1-v3`: 4.6280.

Case	Split	L	T	$D = c^2$	A	β
`wave2d-g0-v0`	train	1	0.5	0.25	0.3	2.5
`wave2d-g0-v1`	train	1.2	0.5	0.5625	0.4	2.5
`wave2d-g0-v2`	train	1	0.75	1	0.5	2.5
`wave2d-g0-v3`	train	1.2	0.75	1.562	0.6	2.5
`wave2d-g1-v0`	validation	1	0.5	0.25	0.3	1.5
`wave2d-g1-v1`	validation	1.2	0.5	0.5625	0.4	1.5
`wave2d-g1-v2`	validation	1	0.75	1	0.5	1.5
`wave2d-g1-v3`	validation	1.2	0.75	1.562	0.6	1.5

Reserved cases: 4 (parameters kept evaluator-only).

3 Group FPDE examples (16 public, 4 reserved cases)

All start from rest and use $t \in (0, 1]$. The fractional PDEs use $x \in (0, \pi)$ with $u(0, t) = u(\pi, t) = 0$ and $u(x, 0) = 0$. The exact solutions are $u = t^p$ (FODE) and $u = t^p \sin x$ (FPDEs). The paper instances are $p = 3$ with order coefficient 0.5 (so $u = t^3$ or $u = t^3 \sin x$); the other cases vary the order and p.

3.1 Fractional ODE

$$ {}_0^C D_t^\alpha u(t) = f(t), \quad u(0) = 0, \qquad u(t) = t^p, \qquad f(t) = \frac{\Gamma(p+1)}{\Gamma(p+1-\alpha)}\, t^{p-\alpha}. $$

Case	Split	α	p
`fode-g0-v0`	train	0.3	2.5
`fode-g0-v1`	train	0.7	3.5
`fode-g1-v0`	validation	0.5	3
`fode-g1-v1`	validation	0.9	4

Reserved cases: 1 (parameters kept evaluator-only).

3.2 Caputo FPDE

$$ {}_0^C D_t^\alpha u = u_{xx} + f(x, t), \qquad u = t^p \sin x, \qquad f(x, t) = \Big[\frac{\Gamma(p+1)}{\Gamma(p+1-\alpha)}\, t^{p-\alpha} + t^p\Big] \sin x. $$

Case	Split	α	p
`fpde_caputo-g0-v0`	train	0.3	2.5
`fpde_caputo-g0-v1`	train	0.7	3.5
`fpde_caputo-g1-v0`	validation	0.5	3
`fpde_caputo-g1-v1`	validation	0.9	4

Reserved cases: 1 (parameters kept evaluator-only).

3.3 Time-variable-order FPDE

$$
{}_0^C D_t^{\alpha(t)} u = u_{xx} + f(x,t), \qquad \alpha(t) = c\,\sin^2 t, \qquad u = t^p \sin x,
$$
$$
f = \Big[\frac{\Gamma(p+1)}{\Gamma(p+1-\alpha(t))}\, t^{p-\alpha(t)} + t^p\Big] \sin x.
$$

Case	Split	c	p
`fpde_variable_order-g0-v0`	train	0.3	2.5
`fpde_variable_order-g0-v1`	train	0.7	3.5
`fpde_variable_order-g1-v0`	validation	0.5	3
`fpde_variable_order-g1-v1`	validation	0.9	4

Reserved cases: 1 (parameters kept evaluator-only).

3.4 Space–time-variable-order FPDE

$$
{}_0^C D_t^{\alpha(x,t)} u = u_{xx} + f(x,t), \qquad \alpha(x,t) = c\,t^2 \sin x, \qquad u = t^p \sin x,
$$
$$
f = \Big[\frac{\Gamma(p+1)}{\Gamma(p+1-\alpha(x,t))}\, t^{p-\alpha(x,t)} + t^p\Big] \sin x.
$$

Case	Split	c	p
`fpde_variable_order_xt-g0-v0`	train	0.3	2.5
`fpde_variable_order_xt-g0-v1`	train	0.7	3.5
`fpde_variable_order_xt-g1-v0`	validation	0.5	3
`fpde_variable_order_xt-g1-v1`	validation	0.9	4

Reserved cases: 1 (parameters kept evaluator-only).